UNIVERSITY OF GHANA

DEPARTMENT OF COMPUTER SCINCE

MSC/ MPHIL DATA SCIENCE

Course Title and Code: ***DSCD 614 Reinforcement learning***

SEMESTER PROJETC TITLE: **Implementation of Proximal Policy Optimization For Energy-Efficient Building Temperature Control:**

Option PPO-2 Energy-Efficient Building Temperature Control

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1. INTRODUCTION AND BACKGROUND STUDY

1.1 Introduction

Buildings must maintain acceptable indoor thermal conditions while using energy efficiently. Heating, ventilation and air-conditioning (HVAC) systems are therefore an important control target because their operation directly affects building energy consumption and occupant comfort. Temperature control is challenging because outdoor conditions, occupancy, electricity prices and building thermal dynamics change over time. Reinforcement learning (RL) has consequently attracted attention for building energy management because it can learn adaptive control policies while balancing competing objectives such as energy efficiency and thermal comfort (Sierla et al., 2022; Al Sayed et al., 2024).

Conventional rule-based thermostats provide simple and interpretable temperature regulation, but their decisions follow predefined rules. They do not learn how present actions influence future thermal conditions or operating costs. RL instead treats control as a sequential decision-making problem in which an agent observes a state, selects an action, receives a reward and transitions to a new state. This formulation is appropriate for HVAC systems because control actions can influence building conditions beyond the immediate time step.

The present study addresses the approved **PPO-2: Energy-Efficient Building Temperature Control** project, which requires heating or cooling to be controlled to minimise energy consumption while maintaining occupant comfort and includes a thermostat-based controller as the baseline. The implemented environment represents the building using seven continuous state variables: indoor temperature, outdoor temperature, comfort setpoint, electricity price, occupancy, and sine/cosine representations of hour of day. The HVAC action is continuous and bounded between **−1 and +1**.

1.2 Research Problem and Significance

The central problem is how to regulate HVAC operation so that thermal comfort and safety are maintained without unnecessary energy consumption or electricity cost. Although a thermostat provides a useful baseline, its fixed control logic does not adapt through experience. A learning-based controller may instead learn how current conditions and actions affect future outcomes.

The significance of this study lies in evaluating this possibility through a controlled and reproducible experiment. The contribution is not the PPO algorithm itself, which is available through an RL library, but the **MDP formulation, custom environment, thermal dynamics, reward design, thermostat baseline, multi-seed training, evaluation framework and analysis** developed for the building-control problem.

1.3 Research Aim

The aim of this study is:

To implement and evaluate a Proximal Policy Optimization-based controller for energy-efficient building temperature regulation within a simulated building environment.

1.4 Research Objectives

The study aims to:

1. Formulate building temperature control as a Markov Decision Process.
2. Implement and validate the custom environment, thermal dynamics and reward function.
3. Train PPO using a reproducible multi-seed protocol.
4. Evaluate PPO against a rule-based thermostat using unseen validation profiles and energy, cost, comfort and safety metrics.
5. Analyse convergence, training stability, exploration and reward-design effects.
6. Examine limitations and challenges associated with real-world HVAC deployment.

1.5 Research Questions

RQ1: How effectively can PPO learn a sequential HVAC control policy that maintains building temperature while reducing energy consumption and electricity cost?

RQ2: How does PPO perform relative to a rule-based thermostat in energy consumption, electricity cost, thermal comfort and safety?

RQ3: How consistent is PPO performance across independent random seeds?

RQ4: How do convergence, exploration and reward design influence the learned temperature-control behaviour?

1.6 Background and Related Work

1.6.1 Reinforcement Learning for Building Control

RL provides a framework for sequential decision-making in which an agent learns through interaction with an environment. Its objective is to maximise cumulative future reward rather than optimise an isolated immediate decision. Building temperature control fits this framework because HVAC actions influence subsequent indoor temperature and future energy requirements. Reviews of RL applications to HVAC systems identify energy, cost and thermal comfort as major control objectives (Sierla et al., 2022).

1.6.2 Conventional HVAC Control and RL

Rule-based thermostats remain attractive because they are simple, interpretable and inexpensive to operate. However, their fixed rules may provide limited adaptability when weather, occupancy or electricity prices change. RL provides an alternative by learning a policy that maps observed building conditions to control actions. This is particularly relevant when energy efficiency and comfort must be optimised simultaneously.

1.6.3 PPO for Continuous Control

Proximal Policy Optimization (PPO) is a policy-gradient algorithm introduced by Schulman et al. (2017). It uses a clipped surrogate objective to limit excessively large policy updates, providing a practical approach to policy optimisation. Schulman et al. (2017)

PPO was selected because the present HVAC problem uses a **continuous one-dimensional action space**. The implementation uses a two-layer policy/value network with architecture **[64, 64]**, allowing the learned policy to control both the direction and magnitude of HVAC intervention.

1.6.4 Reward Design

Reward design is critical because energy efficiency and thermal comfort can conflict. Minimising HVAC energy alone could produce unacceptable temperatures, whereas excessive comfort weighting could increase unnecessary HVAC operation. RL-based HVAC research therefore commonly considers multiple objectives within the reward formulation (Sierla et al., 2022).

The present study combines **energy consumption, electricity cost, thermal discomfort and temperature-limit violations**, explicitly representing the desired energy–comfort–safety trade-off.

1.6.5 Research Gap and Motivation

Despite promising simulation results, transferring RL-based HVAC policies to real buildings remains challenging. Recent reviews identify changing weather and occupancy, limited generalisation, data requirements, safety and the simulation-to-real-world transition as important barriers (Al Sayed et al., 2024).

This study therefore focuses on a **controlled simulation experiment** rather than claiming immediate real-world applicability. Its purpose is to determine whether PPO can learn a reproducible continuous HVAC policy and achieve a better energy–comfort trade-off than the specified thermostat baseline. The experimental design follows the course requirements for multi-seed training, unseen evaluation, baseline comparison and reporting variation across runs.

2. METHODOLOGY

2.1 Problem Formulation

2.1.1 Markov Decision Process

Building temperature regulation was formulated as a Markov Decision Process (MDP),

$$M = (S, A, P, R, \gamma),$$

where $\boldsymbol{S}$ is the state space, $\boldsymbol{A}$ the action space, $\boldsymbol{P}$ the transition dynamics, R the reward function and $\boldsymbol{\gamma}$ the discount factor. This formulation represents HVAC control as sequential decision-making because an action taken at one hour changes the subsequent indoor temperature and therefore influences future energy consumption, comfort and control requirements. This directly follows the course emphasis on modelling real-world problems through states, actions, rewards, transitions and discounting.

2.1.2 State Space

At time *t* the environment provides a seven-dimensional continuous observation:

$$s_t = [\, T_t^{in}, T_t^{out}, T_t^{set}, P_t, O_t, \sin\left(\frac{2\pi h_t}{24}\right), \cos\left(\frac{2\pi h_t}{24}\right)]$$

The variables represent indoor temperature, outdoor temperature, comfort setpoint, electricity price, occupancy and two cyclic time features. Temperature variables are normalised by 40, occupancy is bounded in [0,1] [0,1], and the cyclic features lie in [−1,1] [-1,1]. Sine/cosine encoding avoids treating hours 23 and 0 as distant points despite their temporal continuity.

The representation was selected to provide the policy with the current thermal condition together with the principal exogenous factors influencing future temperature and operating cost. Under the implemented simulator, these variables provide the information required by the transition model, supporting the Markov assumption. Real buildings may contain additional dependencies, such as thermal mass, solar gains and ventilation, which are outside the modelled state.

2.1.3 Action Space

The controller uses a one-dimensional continuous action,

$$\alpha_t \in [-1, 1],$$

where positive values represent heating, negative values represent cooling and zero represents no HVAC intervention. The continuous formulation allows PPO to learn both the direction and magnitude of control rather than selecting from predefined operating modes.

2.1.4 Transition Dynamics

Following action α_t, indoor temperature is updated by the implemented physics-inspired thermal model.

$$T_{t+1}^{in} = T_t^{in} + \alpha\left(T_t^{out} - T_t^{in}\right) + \beta\alpha_t + \eta O_t + \epsilon_t,$$

Temperature evolution depends on the previous indoor temperature, outdoor temperature, HVAC action and occupancy-related heat gain. The principal configured parameters are $\alpha = 0.08$for outdoor-temperature coupling, $\beta = 3.0$ for the HVAC effect and η=0.15 for occupancy heat gain; the final disturbance standard deviation is zero.

The transition therefore links present actions to future states, establishing the sequential dependency required for RL control.

2.1.5 Reward Function

The reward was designed specifically for this project and combines four objectives:

$$R_t = -(w_E E_t + w_c C_t + w_D D_t + w_V V_t),$$

where E is normalized HVAC energy, C electricity cost, D thermal discomfort and V a severe temperature-violation indicator. The weighting prioritizes thermal comfort and safety while retaining incentives to reduce energy consumption and electricity expenditure.

Discomfort is zero within the permitted comfort band and increases with deviation beyond it, while severe violations receive the strongest penalty. Thus, the reward represents the intended **energy–cost–comfort–safety trade-off** rather than energy minimization alone.

2.1.6 Episode Structure and Discounting

Each operating profile forms a **24-hour episode** with one-hour time steps. No environmental failure condition produces terminated = True; instead, the episode is truncated when the 24-hour horizon is reached. The implementation therefore distinguishes environmental termination from time-limit truncation.

The discount factor is $\gamma = 0.99$. This places substantial weight on future rewards, which is appropriate because HVAC actions affect subsequent indoor temperatures and future control requirements. The finite 24-hour horizon also makes consideration of delayed consequences important.

2.2 Experimental Development

2.2.1 Experimental Design

The experiment followed a controlled simulation workflow comprising **environment construction, PPO training, baseline development, multi-seed validation and comparative evaluation**. The implementation separates the custom environment, thermal model, reward calculator and thermostat baseline, while PPO is provided through Stable-Baselines3. This separation reflects the project's original contribution beyond the library algorithm.

The final PPO experiment used **three independent seeds: 42, 123 and 2024**, with the same configuration and **20,224 environment timesteps per seed**. Holding the configuration constant across seeds allows variation in results to be interpreted as training stochasticity rather than configuration differences.

2.2.2 Environment Development

A custom Gymnasium-compatible ***BuildingTemperatureEnv*** was developed to represent a single-zone building. The environment receives hourly operating profiles, maintains the current indoor temperature and simulation time, and returns the seven-dimensional observation after each action.

At every step, PPO supplies a continuous HVAC action, the thermal model updates indoor temperature, and the reward calculator evaluates energy, electricity cost, discomfort and safety. The resulting state and reward are then returned to the agent. Evaluation profiles can be explicitly selected, allowing identical unseen profiles to be used for PPO and the thermostat.

2.2.3 PPO Model and Training Configuration

PPO was implemented using the ***Stable-Baselines3 PPO algorithm with MlpPolicy***. The policy and value networks use two hidden layers of 64 units each.

Table 1: PPO model and training configuration

Hyperparameter	Value
Learning rate	0.0003
Rollout steps	256
Batch size	64
PPO epochs	10
Discount factor γ\gamma	0.99
GAE λ\lambda	0.95
Clip range	0.20
Entropy coefficient	0.01
Value-function coefficient	0.50
Maximum gradient norm	0.50
Network architecture	64, 64

A **5,000-timestep pilot** was first used to assess whether the configuration produced meaningful learning. The final experiment subsequently used approximately 20,000 timesteps, with each run completing 20,224 timesteps because of the PPO rollout structure.

Training returns were recorded using *TensorBoard's* rollout/*ep_rew_mean.* The final analysis extracted actual logged observations rather than reconstructing rewards from evaluation data.

2.2.4 Baseline Development

A rule-based thermostat was implemented as the required baseline. It uses a **22°C setpoint with a 1°C tolerance** and determines HVAC intervention from the current indoor temperature.

For a fair comparison, PPO and the thermostat use the same building environment, thermal dynamics, validation profiles, reward calculations and metric-extraction procedures. This isolates differences in controller behaviour rather than differences in evaluation conditions.

2.2.5 Evaluation Protocol

The final evaluation uses **30 unseen validation profiles**, each representing a 24-hour episode. PPO was evaluated independently for Seeds 42, 123 and 2024, producing **720 hourly observations per seed and 2,160 observations overall**. The thermostat was evaluated on the same 30 profiles, producing 720 observations.

PPO evaluation used deterministic action selection, thereby disabling stochastic exploration during evaluation. The evaluation harness verified profile identity, episode length, action bounds and metric completeness for both controllers.

Performance was assessed using total HVAC energy, electricity cost, mean temperature deviation, total discomfort, comfort-band coverage, temperature-safety violations, cumulative reward and peak HVAC energy. PPO results were aggregated across the three seeds using the mean and standard deviation rather than selecting the strongest run.

2.3 Methodological Integrity

The methodology deliberately separates **training evidence from held-out evaluation evidence**. Training behaviour is assessed from TensorBoard logs, whereas final performance is calculated from unseen validation profiles. The multi-seed design further reduces dependence on a single stochastic training outcome.

The project-specific contributions include the MDP formulation, custom environment, thermal dynamics, reward function, thermostat baseline, evaluation harness, metric computation, multi-seed protocol and analysis. This aligns with the course emphasis on reproducible RL implementation and experimental evaluation.

3. RESULTS AND DISCUSSION

3.1 Training Performance and Convergence

The final PPO experiment used three independent random seeds **42, 123 and 2024** with an identical configuration and **20,224 environment timesteps per seed**. Using multiple seeds allowed the learning behaviour and final performance to be assessed without relying on a single favourable training run. The project evidence confirms that the three final PPO models were independently seeded and completed the same training budget.

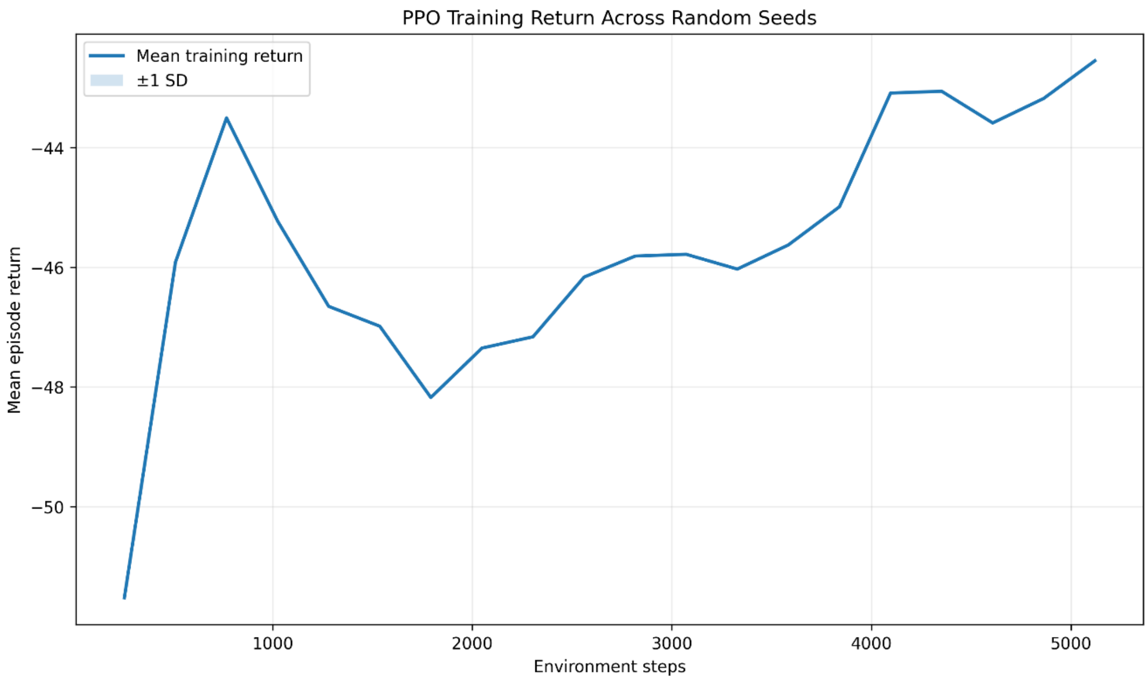

Figure 1. PPO mean training return across the common TensorBoard observation range, with ±1 standard deviation across seeds.

Figure 1 presents the mean training return with ±1 standard deviation over the common recorded training range. The return shows an overall movement from approximately **- 51.5 toward less-negative values**, although the improvement is not monotonic. Temporary declines occur during training, reflecting the stochastic nature of policy optimisation. Nevertheless, the overall upward trend indicates progressive policy improvement.

The training evidence should not be interpreted as proof of global convergence. The experiment used a finite training budget, and the common-range plot is constrained by the available logged observations. It is therefore more appropriate to conclude that PPO demonstrated **substantial learning within the tested training budget**.

The difference between the initial pilot and extended training provides stronger evidence of this improvement. The 5,000-timestep pilot produced only **25.00% comfort-band coverage**, a mean temperature deviation of **1.7385°C**, total discomfort of **202.0544**, and **14 safety violations**. After extended training, these values improved to **99.17%**, **0.3133°C**, **0.2017**, and **zero violations**, respectively. Total energy also decreased from **58.2113 to 37.3308**, while electricity cost decreased from **43.2201 to 28.3042**.

These changes demonstrate that additional interaction with the environment substantially improved the controller's ability to balance energy use and thermal regulation.

3.2 Final Multi-Seed PPO Performance

The final evaluation used **30 unseen 24-hour validation profiles for each seed**, resulting in **90 evaluation episodes and 2,160 hourly observations**. Deterministic action selection was used during evaluation, while the same validation profiles were applied across the PPO runs.

Table 2: Multi-seed PPO performance

Metric	**Seed 42**	**Seed 123**	**Seed 2024**	**Mean ± SD**
Total energy	3.8388	3.9074	3.9063	3.8842 ± 0.039
Electricity cost	2.9559	3.0138	2.9504	2.9734 ± 0.035
Mean temperature deviation	0.3269°C	0.3211°C	0.3300°C	**0**0.3260 ± 0.0045°**C**
Total discomfort	0.0525	0.0190	0.0253	**0.0323 ± 0.0178**
Comfort-band coverage	98.75%	98.89%	99.03%	98.89 ± 0.14%
Safety violations	0	0	0	**0 ± 0**

The results show **moderate variation across seeds**, with energy consumption having a standard deviation of **0.0393** and electricity cost varying by **0.0351**. Mean temperature deviation remains closely grouped across seeds, while comfort-band coverage remains above **98% for every seed**, and all three models record zero safety violations.

This consistency is important because the headline result is based on the **mean performance rather than the best-performing seed**. The three-seed experiment therefore provides evidence that the learned control behaviour is **reasonably consistent and reproducible** under the defined simulation.

3.3 PPO versus the Thermostat Baseline

The rule-based thermostat was evaluated on the **same 30 validation profiles and 24-hour episode structure** as PPO. Profile identities, episode lengths and metric definitions were explicitly verified, providing a controlled basis for comparison.

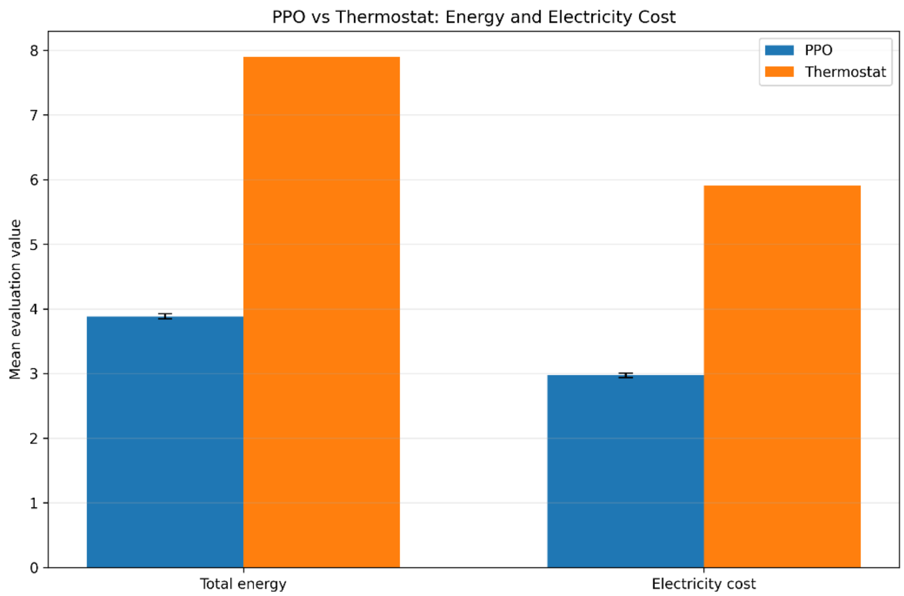

Figure 2. PPO versus thermostat: energy consumption and electricity cost.

PPO achieved mean total energy consumption of **3.7617**, compared with **7.9000** for the thermostat, corresponding to a **52.38% reduction**. Electricity cost decreased from **5.9100 to 2.9240**, a **50.53% reduction**. Peak HVAC energy was also lower under PPO, at **0.3393 compared with 1.0000** for the thermostat.

The efficiency improvement is particularly significant because it did not occur at the expense of thermal comfort.

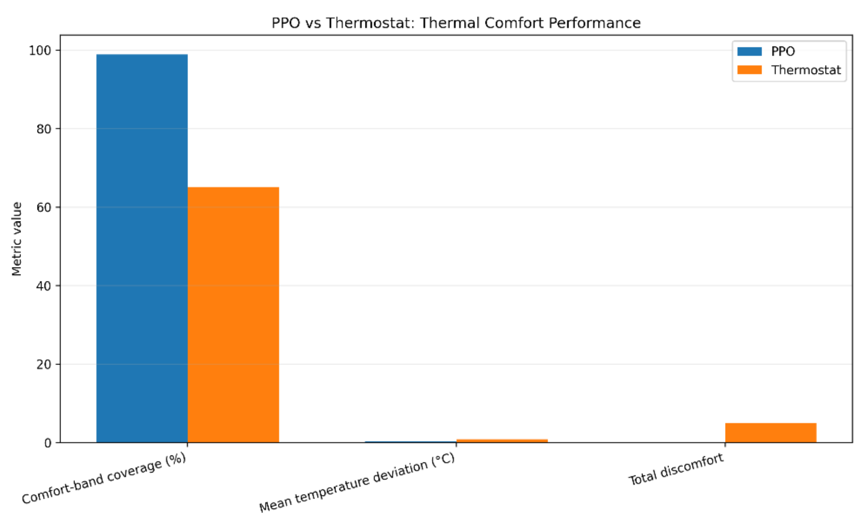

Figure 3. PPO versus thermostat: thermal comfort performance.

PPO achieved **98.33% comfort-band coverage**, compared with **65.14%** for the thermostat. Mean temperature deviation decreased from **0.8363°C to 0.3338°C**, while total discomfort decreased from **4.9176 to 0.0667**, representing a **98.64% reduction**. Both controllers recorded zero safety violations.

Thus, PPO achieved simultaneous improvements in efficiency and comfort. The zero safety violations should nevertheless be interpreted carefully: PPO was **not safer than the thermostat according to this metric**, because both controllers achieved the same value.

3.4 Representative Control Behaviour

The aggregate metrics are supported by the representative 24-hour trajectory of the Seed 123 policy.

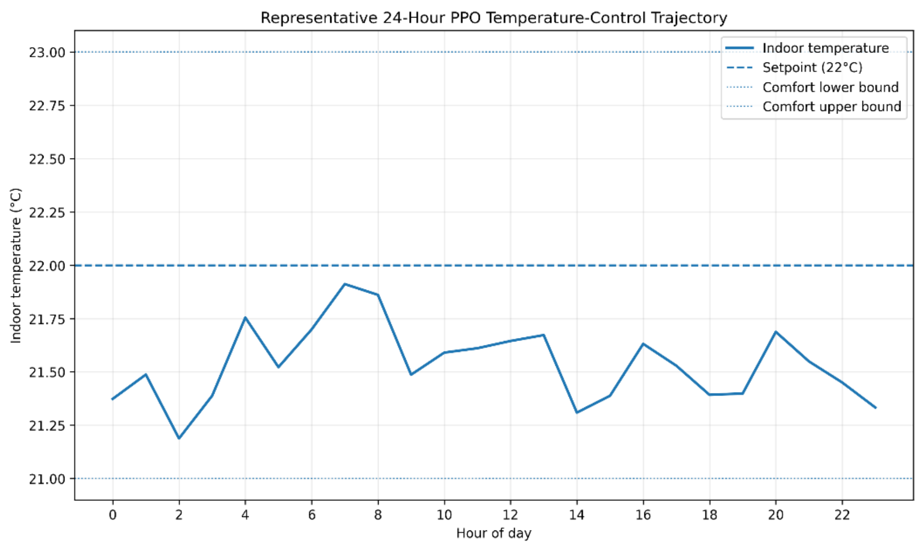

Figure 4. Representative 24-hour PPO temperature-control trajectory.

The indoor temperature remains within the **21–23°C comfort band** throughout the representative day and remains close to the 22°C setpoint. The controller therefore does not continuously force exact setpoint tracking. Instead, it permits small deviations while avoiding unnecessary HVAC intervention. This behavior is consistent with the project's reward design, which balances comfort against energy and cost.

The corresponding control actions provide further insight.

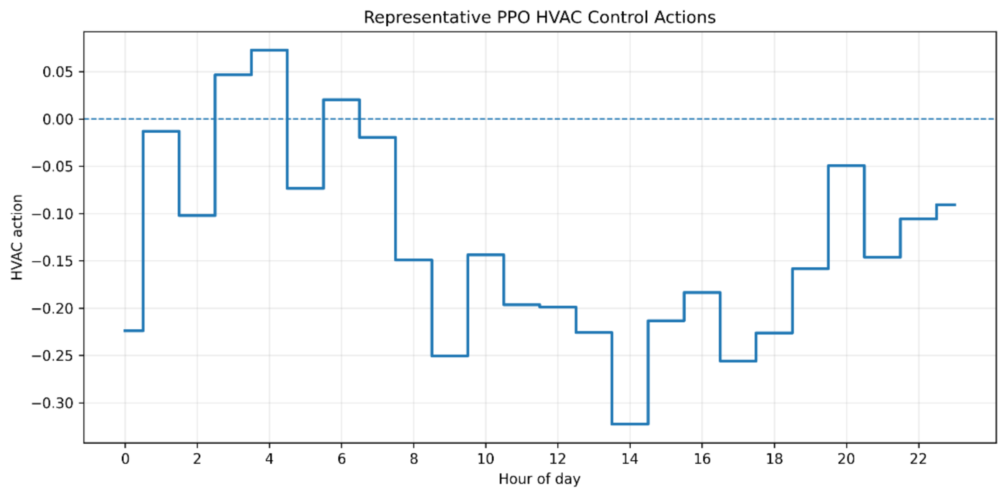

Figure 5. Representative PPO HVAC control actions over 24 hours.

The PPO controller generally applies relatively small continuous actions, varying between heating and cooling as environmental conditions change. This differs from the stronger switching behavior of the thermostat. The learned policy therefore appears to exploit the building's thermal dynamics by making smaller interventions while maintaining acceptable indoor conditions.

These two figures are illustrative rather than substitutes for the aggregate evaluation. The multi-seed statistics remain the principal evidence for overall performance.

3.5 Discussion of Learning Behavior

The results indicate that PPO learned a substantially improved control policy as training progressed. The large improvement between the 5,000-timestep pilot and the extended experiment suggests that the initial training period was insufficient for the agent to discover an effective energy–comfort strategy.

Training stability is also supported by the final multi-seed results. All three seeds produced zero safety violations and comfort coverage above 97%, while energy and electricity-cost variation remained small. However, three seeds provide limited statistical evidence; therefore, the findings support reproducibility under the tested conditions, rather than universal superiority of PPO.

Exploration was important during training because the agent needed to discover how continuous HVAC actions affected future indoor temperature and cumulative reward. During final evaluation, exploration was disabled through deterministic policy execution. This ensures that the reported validation performance represents the learned policy rather than additional stochastic exploration.

The reward design also appears to have influenced the observed behavior appropriately. Because energy, electricity cost, discomfort and severe temperature violations were incorporated simultaneously, the policy was encouraged to reduce HVAC effort without sacrificing thermal conditions. The final PPO results - 52.38% lower energy, 50.53% lower electricity cost and 98.64% lower discomfort than the thermostat—are consistent with this multi-objective formulation.

3.6 Limitations and Deployment Challenges

The results are subject to several important limitations. First, the experiment uses a **simplified single-zone simulated building**. Real buildings involve multiple thermal zones, solar gains, ventilation, thermal mass, equipment dynamics and unpredictable occupant behaviour. Consequently, the observed improvement cannot be assumed to transfer directly to physical HVAC systems.

Second, the evaluation consists of **30 unseen 24-hour profiles**. This is a stronger test than evaluating on training episodes, but it remains a limited representation of long-term building operation. Seasonal weather, unusual occupancy, sensor uncertainty and changing electricity prices may produce conditions outside the validation distribution.

Third, the thermostat is a simple rule-based baseline. The results establish PPO's advantage over the implemented thermostat, but not over every conventional HVAC-control strategy. Comparisons with model-predictive control or alternative RL methods would provide stronger evidence of relative performance.

Finally, physical deployment introduces safety concerns absent from the simulation. Sensors may fail or produce noisy measurements, actuators have physical limits and delays, and learned policies may encounter states not represented during training. A practical deployment should therefore use **supervisory safety constraints and a conventional fallback controller**, with PPO initially operating in a controlled or advisory capacity.

Overall, the experiment demonstrates that PPO learned an effective continuous HVAC control policy within the defined simulation. Extended training substantially improved comfort and safety relative to the initial pilot, while the final three-seed evaluation showed consistent behaviour across independent runs.

Most importantly, PPO achieved a substantially better energy–comfort trade-off **than the thermostat:** 52.38% lower energy consumption, 50.53% lower electricity cost, 60.09% lower mean temperature deviation, 98.64% lower discomfort, and 98.33% comfort-band coverage**.** Both controllers recorded zero safety violations.

The findings therefore support the effectiveness of PPO **under the project's simulated conditions**, while the limitations identified above prevent the results from being interpreted as direct evidence of real-world deployment readiness.

4. CONCLUSION AND FURTHER WORK

4.1 Conclusion

This study implemented and evaluated a **Proximal Policy Optimization (PPO)** controller for sequential building temperature control. The problem was formulated as a Markov Decision Process in which the agent observes building operating conditions, selects a continuous HVAC action, and receives a reward balancing energy consumption, electricity cost, thermal comfort and safety.

The results demonstrate that PPO learned an effective control policy within the simulated environment. Extended training increased comfort-band coverage from **25.00% to 99.17%** and reduced safety violations from **14 to zero**. The final evaluation across Seeds **42, 123 and 2024** also showed consistent performance, with all seeds achieving zero safety violations and comfort-band coverage above 97%.

Compared with the rule-based thermostat, PPO achieved a substantially better energy–comfort trade-off: **52.38% lower energy consumption, 50.53% lower electricity cost and 98.64% lower total discomfort**, while achieving **98.33% comfort-band coverage** compared with 65.14% for the thermostat. These findings indicate that PPO can exploit sequential thermal dynamics to coordinate HVAC actions more efficiently than the implemented threshold-based controller.

However, these findings are bounded by the simulated environment, validation profiles and selected baseline. They demonstrate effectiveness and reproducibility under the defined experimental conditions, but do not establish equivalent performance in physical buildings.

4.2 Further Work

Future work should extend validation to **longer and more diverse operating periods**, including seasonal weather, realistic occupancy patterns and measured electricity-price data. The environment could also be expanded to **multiple thermal zones** and more realistic HVAC equipment constraints.

Further experiments should compare PPO with **model-predictive control, optimisation-based controllers and alternative RL algorithms** to establish whether the observed improvement is specific to PPO.

Finally, deployment research should investigate **safe RL and supervisory control**, incorporating explicit temperature, actuator and equipment constraints. A staged pathway from simulation to offline validation and advisory operation would provide a safer basis for eventual integration with real building-management systems.

AI Use Justification and Declaration

Artificial Intelligence (AI), specifically ChatGPT, was used as a support tool during the development and documentation of this project. Its use was limited to assisting with understanding reinforcement-learning concepts, structuring the methodology, reviewing implementation logic, interpreting experimentally generated results, and improving the clarity and organization of the written report. AI was not used to fabricate experimental results, generate or alter reported performance values, or replace the project's actual implementation and evaluation process**.** All reported results, figures and conclusions were derived from experiments conducted within the project environment and subsequently verified by the project team. The final content was reviewed and validated before used. Below were some prompts used:

- Help formulate the building temperature-control problem as a Markov Decision Process, clearly defining the state space, continuous action space, transition dynamics, reward function and termination conditions based on the implemented project environment.
- Review the PPO training and evaluation workflow for methodological consistency. Identify potential issues in reproducibility, multi-seed evaluation, validation-profile separation, action bounds and metric calculation, and suggest verification checks without inventing results.
- Using only the experimentally generated PPO and thermostat evaluation results provided, identify the main findings concerning energy consumption, electricity cost, thermal comfort, discomfort and safety. Explain what the results demonstrate without overstating the conclusions.

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